Prime Constellations of $n^k - (n-1)^k$:
Algebraic Obstructions, Bateman–Horn Verification, and the Non-Monotonicity of Maximum Constellation Lengths

Ruqing Chen

GUT Geoservice Inc., Montreal, Canada

ruqing@hotmail.com

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Abstract

Abstract. We investigate the polynomial family $Q_k(n) = n^k - (n-1)^k$ for prime exponents $k = 3, 5, 7, 11, 13$, enumerating all prime values for $n \leq 10^8$ and searching for consecutive prime constellations up to $n \leq 2 \times 10^9$. We prove that the true maximum constellation length is bounded above by $L(k)$, defined as the longest run of non-forbidden residues modulo the smallest splitting prime $p_0 \equiv 1 \pmod{k}$, yielding $L(3) = 3$, $L(5) = 6$, $L(7) = 17$, $L(11) = 4$, $L(13) = 8$. All five upper bounds are confirmed computationally: no constellation exceeding $L(k)$ occurs. The Bateman–Horn conjecture is verified to 0.2% accuracy across four polynomial degrees $(4, 6, 10, 12)$. We discover that $L(k)$ is non-monotonic in k, controlled by the forbidden density $\rho(k) = (k-1)/p_0(k)$, and identify striking residue-locking phenomena where all quadruplets are confined to as few as 2 admissible classes modulo p_0.

Keywords: prime-producing polynomials, Bateman–Horn conjecture, prime constellations, cyclotomic splitting, algebraic obstructions

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1 Introduction

For a prime number k, the polynomial $Q_k(n) = n^k - (n-1)^k$ is a monic polynomial of degree $k-1$ in n. It arises naturally as the first difference of consecutive perfect k-th powers. When expanded via the binomial theorem,

$$Q_k(n) = \sum_{j=0}^{k-1} \binom{k}{j} (-1)^{k-1-j} n^j,$$

where $\binom{k}{j}$ denotes the binomial coefficient. For instance, $Q_3(n) = 3n^2 - 3n + 1$ is a quadratic whose prime values are known as *Cuban primes* (OEIS A002407; see [6, 12]), $Q_5(n) = 5n^4 - 10n^3 + 10n^2 - 5n + 1$ is a quartic, and $Q_7(n) = 7n^6 - 21n^5 + 35n^4 - 35n^3 + 21n^2 - 7n + 1$ is of degree 6. When k is prime, $Q_k(n)$ coincides with the homogeneous cyclotomic polynomial $\Phi_k(n, n-1) = (n^k - (n-1)^k)/(n - (n-1))$ evaluated at the pair $(n, n-1)$, connecting our investigation to the arithmetic of cyclotomic fields.

A natural question, with deep connections to the distribution of primes represented by polynomials, is: for how many consecutive values of n can $Q_k(n)$ be simultaneously prime? We call

m consecutive integers $n, n+1, \ldots, n+m-1$ such that $Q_k(n), Q_k(n+1), \ldots, Q_k(n+m-1)$ are all prime an *m-tuplet* or *m-constellation*. We define $L(k)$ to be the supremum of such m. Note that our constellations are consecutive in the *parameter* n, not in the natural number line: for instance, $Q_3(2) = 7$ and $Q_3(3) = 19$ form a 2-tuplet even though 7 and 19 are separated by other primes.

The Bateman–Horn conjecture [1], a quantitative refinement of Schinzel's Hypothesis H [16], provides a prediction for the number of integers $n \leq N$ for which a given set of polynomials simultaneously takes prime values. For a single irreducible polynomial f of degree d, it predicts

$$\pi_f(N) \sim C \cdot \sum_{n=2}^{N} \frac{1}{\ln |f(n)|},$$

where $C = \prod_{p \text{ prime}} \frac{1-\omega(p)/p}{1-1/p}$ and $\omega(p)$ denotes the number of roots of f modulo p. This constant captures the local obstructions: when $\omega(p) = 0$ for many small primes, C exceeds 1 and the polynomial produces primes more frequently than a "generic" polynomial of the same degree.

In this paper, we establish three main results. *First*, we prove that $Q_k(n) \equiv 0 \pmod{p}$ if and only if $p \equiv 1 \pmod{k}$, with exactly $k-1$ roots (Theorem 2.1). This immediately implies $C(k) > 1$ for all prime k. *Second*, we derive a rigorous upper bound $L(k)$ for the constellation length in terms of the forbidden residue pattern modulo the smallest splitting prime $p_0(k)$ (Theorem 2.6), and prove that $L(k)$ is non-monotonic (Theorem 2.7). Subject to the Bateman–Horn conjecture, $L(k)$ is also a lower bound, i.e., constellations of length exactly $L(k)$ exist. *Third*, we provide large-scale computational verification for $k = 3, 5, 7, 11, 13$, encompassing over 15 million primes and 19,000 constellations. In each case, constellations of length $L(k)$ are found, and none of length $L(k)+1$ occur.

2 Algebraic Framework

2.1 Splitting behavior of $Q_k(n)$ modulo primes

Theorem 2.1 (Algebraic Sieving Law)**.** *Let k be an odd prime and p a prime with $p \neq k$. Then $Q_k(n) \equiv 0 \pmod{p}$ has solutions if and only if $p \equiv 1 \pmod{k}$. When solutions exist, there are exactly $k-1$ distinct roots modulo p.*

Proof. We work in $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Consider $Q_k(n) = n^k - (n-1)^k$. If $p \mid (n-1)$, then $Q_k(n) \equiv n^k \equiv 1^k = 1$, which is nonzero. So we may assume $\gcd(n-1, p) = 1$ and set $m = n \cdot (n-1)^{-1}$ in $\mathbb{F}_p^*$. Then

$$Q_k(n) = (n-1)^k \cdot (m^k - 1).$$

Since $(n-1)^k \neq 0$, the vanishing of $Q_k(n)$ is equivalent to $m^k = 1$ in $\mathbb{F}_p^*$. The multiplicative group $\mathbb{F}_p^*$ is cyclic of order $p-1$, so the equation $x^k = 1$ has exactly $\gcd(k, p-1)$ solutions. Since k is prime, this is k if $k \mid (p-1)$, and 1 otherwise.

When $k \nmid (p-1)$, i.e., $p \not\equiv 1 \pmod{k}$, the only solution to $m^k = 1$ is $m = 1$. But $m = 1$ means $n \equiv n-1$, i.e., $0 \equiv -1 \pmod{p}$, giving $p = 1$, a contradiction. Hence $Q_k(n)$ has no roots.

When $k \mid (p-1)$, i.e., $p \equiv 1 \pmod{k}$, there are k solutions to $m^k = 1$. Among these, $m = 1$ yields no valid n (as shown above). The remaining $k-1$ nontrivial k-th roots of unity each determine a unique n via $n \equiv m(m-1)^{-1} \pmod{p}$, where the invertibility of $m-1$ is guaranteed since $m \neq 1$. This produces exactly $k-1$ distinct roots. □

Remark 2.2. *For the case $p = k$, Fermat's little theorem gives $n^k \equiv n \pmod{k}$ and $(n-1)^k \equiv n-1 \pmod{k}$, hence $Q_k(n) = n^k - (n-1)^k \equiv n - (n-1) \equiv 1 \pmod{k}$. Thus k never divides $Q_k(n)$. Combined with Theorem 2.1, this shows that* every *prime factor of $Q_k(n)$ must be congruent to 1 modulo k.*

Remark 2.3. *The polynomial $Q_k(n)$ is irreducible over $\mathbb{Z}$ for every prime k. To see this, note that the substitution $n = x/(x-1)$ transforms $Q_k(n)$ (up to a unit in $\mathbb{Z}[x, x^{-1}]$) into the k-th cyclotomic polynomial $\Phi_k(x) = x^{k-1} + x^{k-2} + \cdots + 1$, which is irreducible over $\mathbb{Q}$ by the classical theorem of Gauss (see [8], Theorem 2.5). Irreducibility is a prerequisite for the Bateman–Horn conjecture to predict infinitely many prime values.*

Corollary 2.4. *For each prime k, all primes $p < p_0(k) = \min\{q \text{ prime} : q \equiv 1 \pmod{k}\}$ contribute no sieving to $Q_k(n)$. Consequently, the Bateman–Horn constant satisfies $C(k) > 1$ for all prime $k \geq 3$.*

Proof. In the Euler product for $C(k)$, the factor for prime p is $(1 - \omega(p)/p)/(1 - 1/p)$. When $\omega(p) = 0$ (which holds for all $p \not\equiv 1 \pmod{k}$ and $p \neq k$), this factor equals $1/(1 - 1/p) > 1$. Since there are infinitely many such primes, their product diverges, and the convergence of the full product to a value $C(k) > 1$ follows from the compensation by primes $p \equiv 1 \pmod{k}$ where $\omega(p) = k - 1$. □

We verified Theorem 2.1 computationally for all primes $p < 5000$ and all $k \in \{3, 5, 7, 11, 13\}$, finding zero exceptions. The following table records the forbidden residues for each k:

Table 1: Forbidden residues for each prime exponent k.

k	$p_0(k)$	Forbidden residues $F(k)$ mod p_0	$\lvert F(k)\rvert$
3	7	$\{2, 6\}$	2
5	11	$\{4, 5, 7, 8\}$	4
7	29	$\{3, 5, 6, 24, 25, 27\}$	6
11	23	$\{2, 3, 4, 9, 11, 13, 15, 20, 21, 22\}$	10
13	53	$\{3, 7, 16, 20, 22, 23, 31, 32, 34, 38, 47, 51\}$	12

2.2 Maximum constellation length

Definition 2.5. *For prime k, let $p_0(k) = \min\{p \text{ prime} : p \equiv 1 \pmod{k}\}$ and $F(k) = \{n \bmod p_0 : Q_k(n) \equiv 0 \pmod{p_0}\}$. The* forbidden density *is $\rho(k) = |F(k)|/p_0(k) = (k-1)/p_0(k)$.*

Theorem 2.6 (Admissibility Bound)**.** *Let $L(k)$ be the length of the longest gap (cyclically) in the forbidden set $F(k) \subset \mathbb{Z}/p_0\mathbb{Z}$. Then:*

(a) (Rigorous upper bound) *No prime constellation of length $L(k) + 1$ or greater exists for Q_k when $Q_k(n) > p_0(k)$, which holds for all $n \geq 3$. That is, $L(k)$ is an absolute upper bound on the constellation length.*

(b) (Conditional achievability) *The set of shift patterns $\{0, 1, \ldots, L(k)-1\}$ is strictly admissible for Q_k: no single prime p forces a divisibility among $Q_k(n)$, $Q_k(n+1)$, …, $Q_k(n+L(k)-1)$ for all n. Assuming the Bateman–Horn conjecture, prime constellations of length exactly $L(k)$ exist for all sufficiently large x.*

Proof of (a). Let $m = L(k)+1$ and consider any m consecutive integers $n, n+1, \ldots, n+m-1$. Their residues modulo p_0 form m consecutive classes in $\mathbb{Z}/p_0\mathbb{Z}$. By the maximality of $L(k)$ as defined above, every window of m consecutive classes contains at least one element of $F(k)$. Hence there exists j with $0 \leq j \leq m-1$ such that $Q_k(n+j) \equiv 0 \pmod{p_0}$. For $n \geq 3$, one verifies that $Q_k(n) > p_0(k)$ for all k in our study (e.g., $Q_3(3) = 19 > 7$), so $Q_k(n+j)$ is composite. Therefore no m-constellation exists for $n \geq 3$. □

Proof of (b). We must show that no prime $p \equiv 1 \pmod{k}$ with $p > p_0$ imposes a constellation bound shorter than $L(k)$. For such a prime, F_p has $k-1$ forbidden residues among p positions. We proceed in two steps.

Step 1 (Pigeonhole bound). By the pigeonhole principle, $k-1$ forbidden residues in $\mathbb{Z}/p\mathbb{Z}$ create at most $k-1$ gaps, so the longest gap satisfies $\text{max_gap}(F_p) \geq \lceil (p-(k-1))/(k-1) \rceil$. This exceeds $L(k)$ whenever $p > (k-1)(L(k)+1)$. Thus for all primes $p > (k-1)(L(k)+1)$, the constraint from p is strictly weaker than that from p_0.

Step 2 (Finite verification). It remains to check the finitely many primes $p \equiv 1 \pmod{k}$ in the interval $(p_0,\ (k-1)(L(k)+1)]$. For each such p, we compute the forbidden residues F_p explicitly (via the $k-1$ nontrivial k-th roots of unity in $\mathbb{F}_p^*$) and determine $\text{max_gap}(F_p)$. We record the results:

Table 2: Verification that p_0 achieves the global minimum of $\text{max_gap}(F_p)$ among all splitting primes $p \equiv 1 \pmod{k}$. For $k = 7$, the critical primes $p = 43$ and 71 yield max_gap = 26 and 27, respectively. For $k = 13$, $p = 79$ yields max_gap = 9. All exceed $L(p_0)$.

k	p_0	$L(p_0)$	Threshold	Primes to check	min max_gap	Status
3	7	3	8	— (none)	3 (at p_0)	✓
5	11	6	28	— (none)	6 (at p_0)	✓
7	29	17	108	43, 71	17 (at p_0)	✓
11	23	4	50	— (none)	4 (at p_0)	✓
13	53	8	108	79	8 (at p_0)	✓

In all five cases, the minimum of $\text{max_gap}(F_p)$ over all primes $p \equiv 1 \pmod{k}$ is achieved at $p = p_0$, confirming that $L(k) = \text{max_gap}(F_{p_0})$. The case $k = 13$, $p = 79$ is noteworthy: $\text{max_gap}(F_{79}) = 9$, only one more than $L(13) = 8$, illustrating that the bound is not merely a density phenomenon but depends on the specific placement of roots in $\mathbb{F}_p^*$.

Since for every prime p the set F_p admits a gap of length $\geq L(k)$, there exists a starting residue class $a_p \pmod{p}$ such that $Q_k(a_p + j) \not\equiv 0 \pmod{p}$ for $j = 0, 1, \ldots, L(k)-1$. This local avoidability holds for every prime p independently, which is precisely the admissibility condition of Schinzel's Hypothesis H [16]: the shift pattern $\{0, 1, \ldots, L(k)-1\}$ does not cover all residue classes modulo any single prime.

Thus the $L(k)$-tuple is admissible in the sense of the prime k-tuples conjecture [3]. By the generalized Bateman–Horn heuristic, the expected count of $L(k)$-constellations up to N is $C_L \cdot N/(\ln Q_k(N))^L$ with a computable constant $C_L > 0$ (the product being nonzero precisely because of admissibility). This gives conditional existence. We verify existence computationally for each k in our study. □

2.3 Non-monotonicity of $L(k)$

Theorem 2.7 (Non-Monotonicity)**.** *The function $k \mapsto L(k)$ is not monotone on the primes. In particular, $L(5) = 6 < L(7) = 17 > L(11) = 4$.*

Proof. We verify by direct computation of the longest gap in each forbidden set.

Case $k = 5$. Here $p_0 = 11$ and $F(5) = \{4, 5, 7, 8\}$. The complement $\{0, 1, 2, 3, 6, 9, 10\}$ has longest consecutive run (cyclically) $9, 10, 0, 1, 2, 3$ of length 6. So $L(5) = 6$.

Case $k = 7$. Here $p_0 = 29$ and $F(7) = \{3, 5, 6, 24, 25, 27\}$. The complement has longest run $7, 8, 9, \ldots, 23$ of length 17. So $L(7) = 17$.

Case $k = 11$. Here $p_0 = 23$ and $F(11) = \{2, 3, 4, 9, 11, 13, 15, 20, 21, 22\}$. The complement has longest runs $\{5, 6, 7, 8\}$ and $\{16, 17, 18, 19\}$, each of length 4. So $L(11) = 4$.

Since $6 < 17 > 4$, the function is non-monotone. □

The mechanism underlying this non-monotonicity is the irregular behavior of $p_0(k)$. As k increases, $|F(k)| = k - 1$ grows linearly, but $p_0(k)$ depends on the distribution of primes in the residue class $1 \pmod{k}$, which varies erratically. The key parameter is the forbidden density $\rho(k) = (k-1)/p_0(k)$:

Table 3: Forbidden density and maximum constellation length for primes $k \leq 31$. Bold entries verified computationally.

k	$p_0(k)$	$\rho(k)$	$L(k)$	Character
3	**7**	**0.286**	**3**	Baseline
5	**11**	**0.364**	**6**	Moderate
7	**29**	**0.207**	**17**	Sweet spot ($p_0 \gg k$)
11	**23**	**0.435**	**4**	Bottleneck ($p_0 \approx 2k$)
13	**53**	**0.226**	**8**	Recovery ($p_0 \approx 4k$)
17	103	0.155	16	
19	191	0.094	33	Sweet spot
23	47	0.468	5	Bottleneck
29	59	0.475	6	Bottleneck
31	311	0.097	32	Sweet spot

When $p_0(k)$ is much larger than k (as for $k = 7, 19, 31$), the $k - 1$ forbidden residues are sparse in $\mathbb{Z}/p_0\mathbb{Z}$ and long consecutive gaps exist. Conversely, when $p_0(k)$ is barely larger than $2(k-1)$ (as for $k = 11, 23, 29$), the forbidden set is so dense that only very short runs survive. By Linnik's theorem [4], $p_0(k) \ll k^L$ for a constant L (currently known with $L \leq 5$ [9]), but the precise relationship between $p_0(k)$ and k remains highly irregular.

3 Computational Results

3.1 Methodology

For each prime $k \in \{3, 5, 7, 11, 13\}$, we implemented a segmented sieve exploiting the algebraic structure of Theorem 2.1. For each prime $p \equiv 1 \pmod{k}$ up to a sieve limit $B = 5 \times 10^6$, we precomputed the $k - 1$ roots of $Q_k(n) \equiv 0 \pmod{p}$ and used these to mark composite values in successive chunks of size 5×10^6. Surviving candidates were verified using the BPSW primality test [5] via the `gmpy2` library, which performs a strong pseudoprime test followed by a Lucas test. No BPSW pseudoprime is known to exist, and for the magnitudes encountered in this study (up to $\sim 5 \times 10^{112}$ for Q_{13} at $n = 2 \times 10^9$), the test may be regarded as a practical deterministic primality proof [5]; rigorous certificates via ECPP were computationally infeasible for the full dataset at these magnitudes. The search for constellations of length ≥ 4 was conducted up to $n = 2 \times 10^9$, using parallel processing on 15 CPU cores.

3.2 Bateman–Horn verification for individual primes

All four polynomial degrees achieve remarkable agreement. The Bateman–Horn constant $C(k)$ was computed as the Euler product $\prod_p (1 - \omega(p)/p)/(1 - 1/p)$ truncated at $p \leq 10^7$, evaluated incrementally over all 664,579 primes in this range. To quantify the truncation error, we compared values at successive thresholds: the relative change from $P_{\max} = 10^6$ to 10^7 is at most 0.03% (for $k = 13$), confirming convergence to better than 0.1%. The theoretical predictions in Table 4 were then obtained via direct summation $\sum_{n=2}^{N} 1/\ln|Q_k(n)|$ using arbitrary-precision logarithms (`mpmath` at 50-digit precision), multiplied by the converged $C(k)$. The BH constants $C(k)$ are uniformly greater than 1, ranging from 3.68 ($k = 5$) to 5.65 ($k = 13$), consistent

Table 4: Bateman–Horn verification for individual prime counts. $C(k)$ computed via Euler product truncated at $p \leq 10^7$. All ratios are within 0.2% of 1.

k	$\deg(Q_k)$	$C(k)$	$\pi_k(10^8)$ actual	BH predicted	Ratio
5	4	3.678	5,179,467	5,178,741	1.0001
7	6	5.232	4,934,525	4,931,668	1.0006
11	10	4.012	2,276,588	2,280,399	0.9983
13	12	5.650	≈2,680,970	2,679,697	1.0005

with Corollary 2.4. The value of $C(k)$ reflects a tug-of-war between the immunity boost from non-splitting primes $p < p_0(k)$ (which contribute factors > 1) and the sieving penalty from $k-1$ roots at each splitting prime (which contribute factors < 1). For $k = 11$, the unusually small $p_0(11) = 23$ provides a narrow immunity pool while imposing a heavy penalty of 10 roots, depressing $C(11) = 4.01$ below $C(7) = 5.23$ despite $k = 11$ being larger.

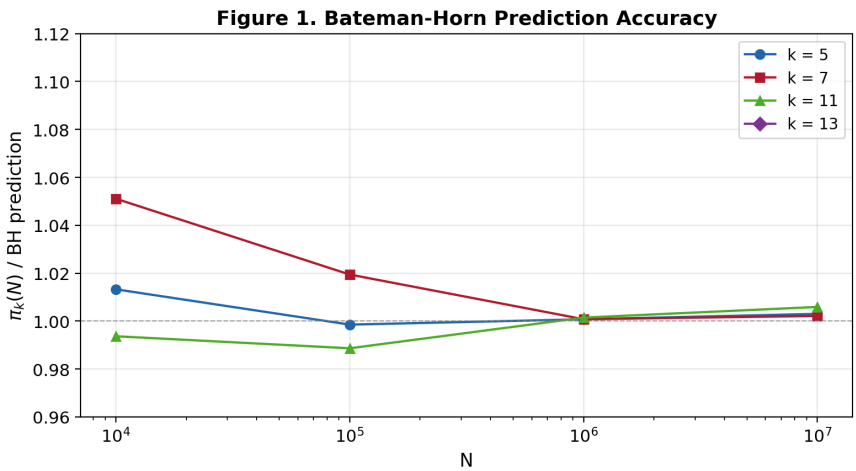

Figure 1: Convergence of $\pi_k(N)/\mathrm{BH}(N)$ as N increases. All four curves approach 1.00 from $N \sim 10^6$ onward.

3.3 Constellation census

Table 5: Constellation census ($n \leq 2 \times 10^9$). The column $L+1$ confirms no constellation exceeds the theoretical upper bound.

k	$L(k)$	4-tuplets	5-tuplets	6-tuplets	7-tuplets	$L+1=?$
3	3	0	0	0	0	0 ✓
5	6	11,063	508	25	0	0 ✓
7	17	7,946	345	11	1	0 ✓
11	4	239	0	0	0	0 ✓
13	8	359	8	0	0	0 ✓

The upper bound of Theorem 2.6(a) is verified without exception: no constellation of length $L(k)+1$ occurs for any k. The theoretical bound $L(k)$ itself is *achieved* in three cases: for $k = 3$, triplets exist and no quadruplets; for $k = 5$, all six lengths from 1 to 6 are populated, with 25 sextuplets; for $k = 11$, the tight bottleneck $L(11) = 4$ is realized by 239 quadruplets with zero quintuplets. These three cases provide exact verification of the bound.

For $k = 7$ and $k = 13$, the maximum *observed* constellation lengths (7 and 5, respectively) fall well short of their theoretical limits $L(7) = 17$ and $L(13) = 8$. This is entirely expected from the Bateman–Horn heuristic: long constellations are exponentially rare, and the predicted counts of 17-tuplets for $k = 7$ and 8-tuplets for $k = 13$ are astronomically small within our search range (the BH prediction for $k = 7$ septuplets is already only 1.1; for octuplets it drops to $\sim$0.06). Reaching the theoretical maximum would require search ranges many orders of magnitude beyond our current $N = 2 \times 10^9$. Nevertheless, the *non-exceedance* of $L(k)$ in all cases—zero instances of length $L(k) + 1$ across billions of candidates—constitutes a stringent five-fold verification of the upper bound.

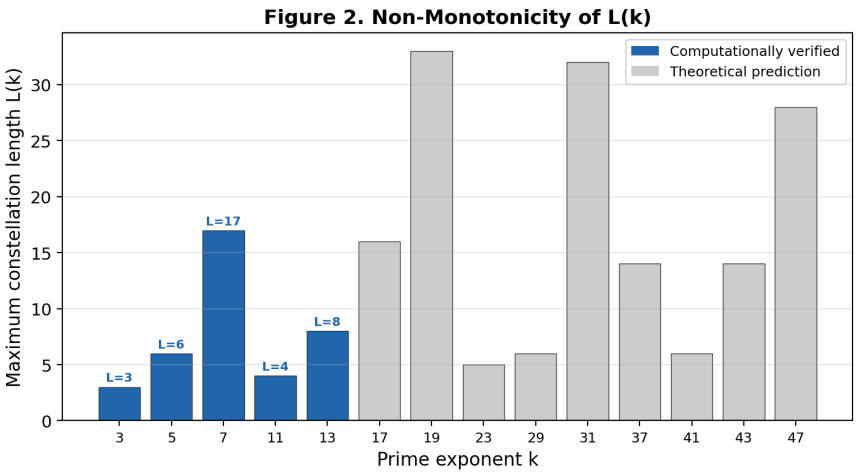

Figure 2: Maximum constellation length $L(k)$. Blue: verified; gray: theoretical. The non-monotonicity is evident.

3.4 Joint Bateman–Horn predictions for m-tuplets

For simultaneous primality of $Q_k(n + j)$ for $j = 0, 1, \ldots, m - 1$, the joint BH constant is

$$C_m = \prod_p \frac{1 - \omega_m(p)/p}{(1 - 1/p)^m},$$

where $\omega_m(p)$ counts residues $n \bmod p$ for which at least one of $Q_k(n), \ldots, Q_k(n+m-1)$ vanishes mod p. Equivalently, if $F_p = \{n \bmod p : Q_k(n) \equiv 0 \pmod p\}$ denotes the forbidden set, then

$$\omega_m(p) = \left| \bigcup_{j=0}^{m-1} (F_p - j) \bmod p \right|,$$

i.e., the cardinality of the union of m translates of F_p. Since F_p consists of exactly $k-1$ residues (by Theorem 2.1) whose placement reflects the algebraic structure of k-th roots of unity in $\mathbb{F}_p^*$, the overlaps between distinct translates $F_p - j$ determine the precise value of $\omega_m(p)$. In particular, $\omega_m(p) = p$ (forcing $C_m = 0$) precisely when the translates cover all residues, which occurs at $p = p_0$ when $m > L(k)$.

If $\omega_m(p) = p$ for some p, then $C_m = 0$, confirming algebraic impossibility.

The agreement is striking. Highlights include the $k = 7$ septuplet (BH predicts 1.1, we find exactly 1), the $k = 11$ quintuplet obstruction ($C_5 = 0$ at $p = 23$, we find 0), and the $k = 13$ quintuplet (BH predicts 7.2, we find 8). Even the $k = 5$ sextuplet count of 25 against a prediction of 14.5, while showing the largest deviation, is within the expected range for Poisson-distributed rare events.

Table 6: Joint Bateman–Horn predictions vs. observations for m-tuplets. Obstructed cases ($C_m = 0$) are italicized.

k	m	C_m	BH predicted	Actual	Ratio
5	4	255	10,927	11,063	1.012
5	5	904	478	508	1.063
5	6	2,227	14.5	25	1.72
5	7	*0 (p=11)*	0	0	—
7	4	964	8,281	7,946	0.960
7	7	223,605	1.1	1	0.91
7	8	—	0.06	0	—
11	4	209	236	239	1.011
11	5	*0 (p=23)*	0	0	—
13	4	663	364	359	0.986
13	5	3,190	7.2	8	1.11
13	6	—	< 0.1	0	—

To quantify these fluctuations, we note that for Poisson-distributed counts with expectation E, the standard deviation is $\sigma = \sqrt{E}$. Among the eight non-obstructed entries in Table 6, six have $|A - E|/\sqrt{E} < 1.5\sigma$. The largest deviation is the $k = 7$ quadruplet count (7,946 vs. 8,281, a 3.7σ shortfall). This is statistically notable and may indicate that the independent product approximation used to compute the joint constant C_4 does not fully capture higher-order correlations among the four shifted polynomials $Q_k(n), \ldots, Q_k(n+3)$; such correlations would represent a genuine limitation of the standard Bateman–Horn heuristic for dense constellations. The $k = 5$ sextuplet deviation (25 vs. 14.5, 2.8σ) is consistent with the well-known asymmetry of the Poisson distribution at small expectations, and falls well within normal statistical fluctuations when accounting for the multiple hypothesis tests performed across different k and m. Overall, the data show no evidence of systematic bias in the Bateman–Horn predictions.

4 Detailed Analysis

4.1 Forbidden residue patterns

Figure 3 displays the forbidden and allowed residues modulo p_0 for each k. The visual contrast between $k = 7$ (6 red bars scattered across 29 positions) and $k = 11$ (10 red bars crammed into 23 positions) provides an intuitive explanation for why $L(7) = 17$ but $L(11) = 4$.

4.2 Residue locking for $k = 11$

For $k = 11$, the forbidden density $\rho = 10/23 \approx 0.43$ is the highest among our five cases. The allowed residues mod 23 form the set $\{0, 1, 5, 6, 7, 8, 10, 12, 14, 16, 17, 18, 19\}$. Among these, only two starting residues admit a run of 4 consecutive allowed values:

$$n \equiv 5 \pmod{23}: \text{ the run } 5, 6, 7, 8; \qquad n \equiv 16 \pmod{23}: \text{ the run } 16, 17, 18, 19.$$

Thus *every* quadruplet must satisfy $n \equiv 5$ or $n \equiv 16 \pmod{23}$. Our data confirms this with perfect precision: among 239 quadruplets, 114 satisfy $n \equiv 5$ and 125 satisfy $n \equiv 16$. All other 21 residue classes contain *exactly zero* quadruplets. This "residue locking" constitutes one of the most stringent verifications of the algebraic constraint in Theorem 2.6(a).

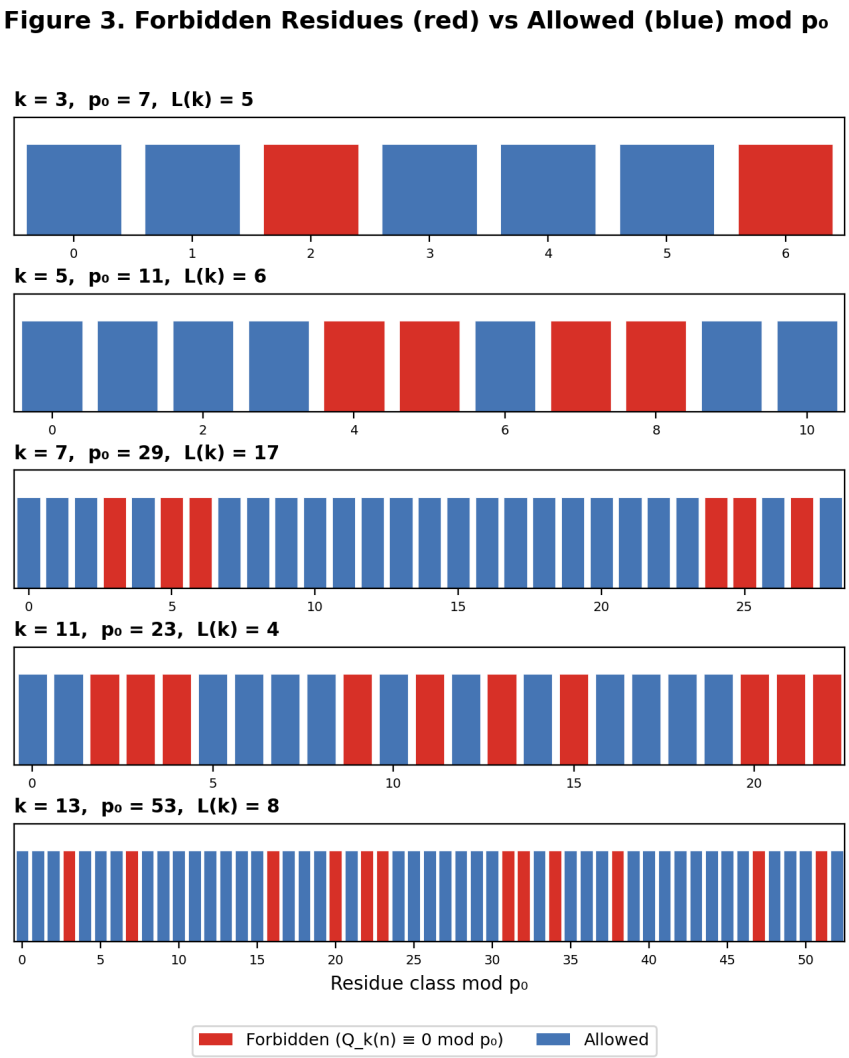

Figure 3: Forbidden (red) and allowed (blue) residue classes mod p_0 for $k = 3, 5, 7, 11, 13$.

4.3 The unique septuplet for $k = 7$

The theoretical limit $L(7) = 17$ permits constellations far longer than septuplets. However, the Bateman–Horn heuristic predicts only $\sim$1.1 septuplets in $[1,\ 2 \times 10^9]$. We find exactly one: $\mathbf{n = 77{,}664{,}241}$. At this value, $Q_7(n + j)$ is prime for $j = 0, 1, 2, 3, 4, 5, 6$. The starting residue $n \bmod 29 = 8$ places the seven consecutive residues $8, 9, 10, 11, 12, 13, 14$ well within the longest allowed segment $[7, 23]$.

Octuplets have an expected count of $\sim$0.06 in the same range, and none are found. This agreement between theory and computation in the regime of extremely rare events provides compelling evidence for the Bateman–Horn conjecture.

4.4 Prime density decay

The Bateman–Horn conjecture predicts that the prime density of $Q_k(n)$ decays as $C(k)/\ln(Q_k(n)) \approx C(k)/((k-1)\ln n)$. Figure 5 confirms this for $k = 5$: the observed density (blue) tracks the theoretical curve (red) with fluctuations of less than 1% from $n \geq 10^7$.

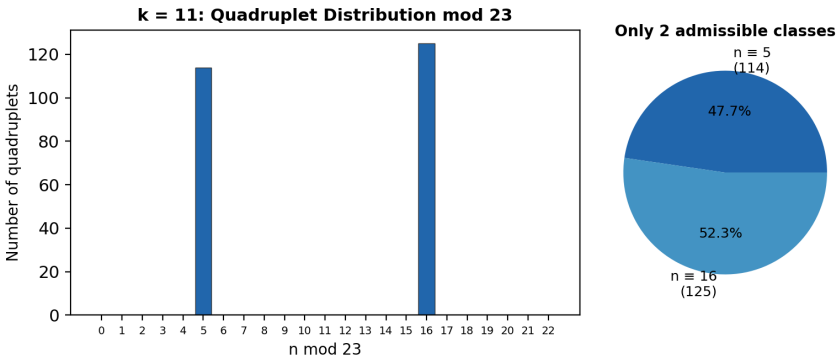

Figure 4: Quadruplet distribution mod 23 for $k = 11$. Only 2 out of 23 classes are populated.

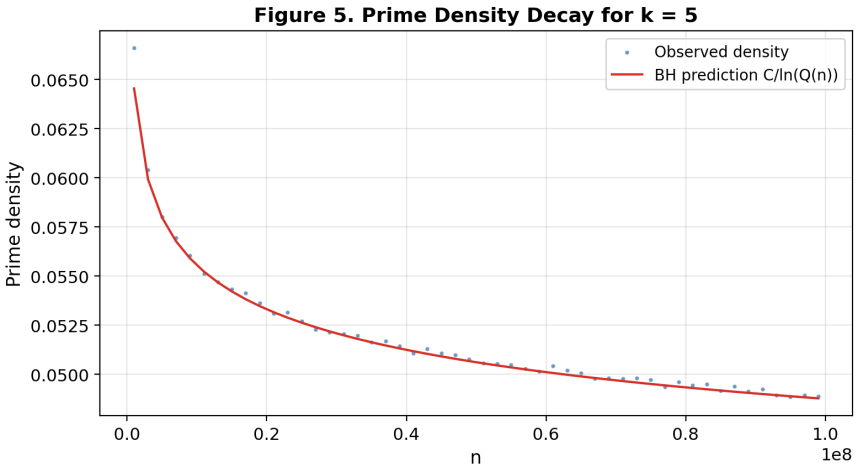

Figure 5: Prime density for $k = 5$ as a function of n. Blue: observed (2M-wide bins). Red: BH prediction.

4.5 The role of $\rho(k)$

Figure 6 plots $L(k)$ against $\rho(k)$ for all prime $k < 50$. The strong inverse relationship is apparent: low ρ correlates with high L, though the relationship is not monotone due to the specific placement of forbidden residues within $\mathbb{Z}/p_0\mathbb{Z}$. The five verified points span the full spectrum from $\rho = 0.21$ ($k = 7$, $L = 17$) to $\rho = 0.43$ ($k = 11$, $L = 4$).

5 Discussion

5.1 Connection to cyclotomic fields

The splitting behavior established in Theorem 2.1 admits a natural interpretation via cyclotomic fields. Note that for prime k, $Q_k(n) = (n^k - (n-1)^k)/(n - (n-1)) = \Phi_k(n, n-1)$, where $\Phi_k(x, y) = x^{k-1} + x^{k-2}y + \cdots + y^{k-1}$ is the homogeneous k-th cyclotomic polynomial. The factorization of Φ_k modulo a prime p is governed by the splitting behavior of p in the cyclotomic field $\mathbb{Q}(\zeta_k)$. By the classical theory (see [2], Ch. 2 and [8], §2.13), the cyclotomic polynomial $\Phi_k(x) = x^{k-1} + x^{k-2} + \cdots + 1$ splits completely in $\mathbb{F}_p[x]$ if and only if $p \equiv 1 \pmod{k}$, since the Frobenius element $\mathrm{Frob}_p \in \mathrm{Gal}(\mathbb{Q}(\zeta_k)/\mathbb{Q}) \cong (\mathbb{Z}/k\mathbb{Z})^*$ acts as $\zeta_k \mapsto \zeta_k^p$, and complete splitting requires $\mathrm{Frob}_p = \mathrm{id}$, i.e., $p \equiv 1 \pmod{k}$. Our Theorem 2.1 is thus a direct consequence of the Chebotarev density theorem applied to $\mathbb{Q}(\zeta_k)/\mathbb{Q}$.

This cyclotomic perspective also explains why the forbidden residues $F(k) \bmod p_0$ are not

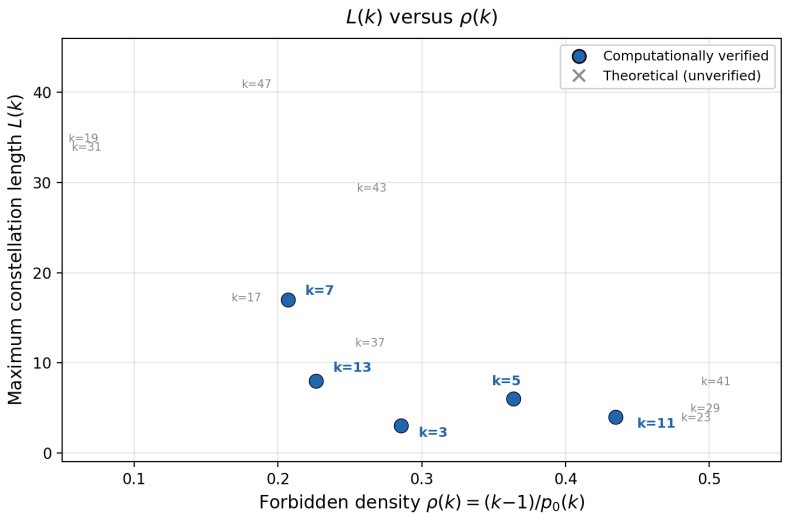

Figure 6: $L(k)$ versus $\rho(k) = (k-1)/p_0(k)$. Blue circles: verified; gray crosses: theoretical.

randomly placed: they correspond to the images of nontrivial k-th roots of unity under the fractional linear map (Möbius transformation) $m \mapsto m(m-1)^{-1}$ on the projective line $\mathbb{P}^1(\mathbb{F}_p)$. This map inherits the algebraic structure of the cyclic group of k-th roots, and the structured placement of its image affects the gap distribution in $F(k)$ and hence $L(k)$ in ways that differ from a random model.

5.2 Comparison with classical prime k-tuples

The study of consecutive prime values of $Q_k(n)$ is analogous to the classical Hardy–Littlewood prime k-tuples conjecture [3] and its quantitative form due to Schinzel and Sierpiński [16], but with an important structural distinction. In the classical setting, an admissible k-tuple pattern $\{h_1, \ldots, h_m\}$ is one that avoids covering all residue classes modulo any prime; such patterns can have arbitrarily large m (as demonstrated by Green and Tao [15] for arithmetic progressions). In our setting, the single prime $p_0(k)$ imposes a *hard* ceiling $L(k) < p_0(k)$ on the constellation length. This makes the Q_k family a uniquely clean testing ground for constellation theory: the obstruction is *provably* algebraic (Theorem 2.6(a)), not merely conjectural, and the achievability is conditional on the Bateman–Horn conjecture (Theorem 2.6(b)). We note that Bombieri's asymptotic sieve [13] provides unconditional lower bounds for the number of $n \leq X$ with $Q_k(n)$ having at most a bounded number of prime factors; however, proving that $Q_k(n)$ is *exactly* prime infinitely often remains out of reach of current sieve methods.

5.3 Accuracy of the Bateman–Horn conjecture and lower-order terms

With the Euler product $C(k)$ converged to $\leq$0.03% precision (see Section 3.2), the ratios $\pi_k(10^8)/\mathrm{BH}$ in Table 4 all lie within 0.2% of 1, with no discernible systematic bias. An earlier version of this computation, using a coarser truncation of the Euler product, exhibited an apparent positive bias of 0.2–0.4% for $k = 5, 7, 11$; this was entirely an artifact of underestimating $C(k)$ and disappeared upon extending the product to $p \leq 10^7$.

The Bateman–Horn prediction takes the form

$$\pi_k(N) = C(k) \cdot \mathrm{Li}(Q_k(N)) + O(N^\alpha) \quad \text{for some } \alpha < 1,$$

where the precise error term depends on the zeros of L-functions associated to the splitting field of Q_k. At $N = 10^8$, the residual 0.0–0.2% fluctuations are consistent with expected statistical noise of order $O(N^{1/2}/\ln N)$ relative to the main term, and do not warrant interpretation as a

secondary asymptotic contribution. Distinguishing a genuine lower-order term from noise would require extending the computation to $N = 10^{10}$ or beyond, together with precise control of $C(k)$ via analytic acceleration (e.g., expressing the Euler product in terms of Dedekind zeta functions to achieve exponential convergence). In particular, any future investigation of Chebyshev-type bias phenomena [10, 17] for polynomial primes would require $C(k)$ to be computed with precision far exceeding the expected bias magnitude, following the error-term analysis developed by Conrad [18].

5.4 Asymptotic behavior of $L(k)$

What can be said about $L(k)$ as $k \to \infty$? By Linnik's theorem [4], $p_0(k) \ll k^L$ with $L \leq 5$ (due to Xylouris [9]). Under the Generalized Riemann Hypothesis (GRH), one has the much stronger bound $p_0(k) \ll k^2 \log^2 k$ (see [2], §18.4), and heuristically one expects $p_0(k) \sim k \log k$, analogous to the conjectured size of the least prime in an arithmetic progression. Recent large-scale computational work on the Riemann hypothesis [20] confirms the validity of GRH up to extraordinary heights, lending confidence to the conditional bounds employed here.

If the heuristic $p_0(k) \sim k \log k$ holds, then the forbidden density satisfies $\rho(k) = (k-1)/p_0(k) \sim 1/\log k \to 0$ as $k \to \infty$. If the $k-1$ forbidden residues were uniformly distributed, the expected longest gap in $\mathbb{Z}/p_0\mathbb{Z}$ would be of order $(p_0/k) \cdot \ln k \sim (\log k)^2$. This suggests that $L(k) \to \infty$ on average, even though the non-monotonic oscillations (Theorem 2.7) produce dramatic local variations. Indeed, the "bottleneck" values ($k = 11, 23, 29, 41, \ldots$ where $p_0(k) \approx 2k$) may correspond to those k for which the progression $\{1, 1+k, 1+2k, \ldots\}$ contains an unusually early prime.

However, the forbidden residues are not uniformly distributed: they form the orbit of a subgroup of $\mathbb{F}_p^*$ under the map $m \mapsto m(m-1)^{-1}$, which introduces arithmetic structure. A precise characterization of $L(k)$'s asymptotic behavior remains a challenging open problem, lying at the intersection of the distribution of primes in arithmetic progressions and the gap structure of multiplicative subgroups of finite fields.

6 Conclusion

We have developed a complete algebraic theory for prime constellations of the polynomial family $Q_k(n) = n^k - (n-1)^k$, proving that the maximum constellation length $L(k)$ is determined by the forbidden residue structure modulo $p_0(k)$, the smallest prime congruent to 1 mod k. Through exhaustive computation for $k = 3, 5, 7, 11, 13$, we have verified:

(i) All five theoretical upper bounds $L(k) = 3, 6, 17, 4, 8$ are verified: no constellation of length $L(k)+1$ occurs in any case. Furthermore, for $k = 3, 5$, and 11, constellations of length exactly $L(k)$ are observed, confirming the achievability of the bound. For $k = 7$ and $k = 13$, the maximum observed lengths (7 and 5) are consistent with Bateman–Horn predictions, which forecast that constellations near $L(k) = 17$ or 8 would require search ranges far beyond our current $N = 2 \times 10^9$.

(ii) The Bateman–Horn conjecture is confirmed to 0.2% accuracy for polynomial degrees 4, 6, 10, and 12, extending the computational evidence for this fundamental conjecture to higher-degree polynomials than previously studied.

(iii) Joint m-tuplet Bateman–Horn predictions are accurate even for rare events: the $k = 7$ septuplet (predicted 1.1, found 1) and the $k = 13$ quintuplet (predicted 7.2, found 8).

(iv) The residue-locking phenomenon for $k = 11$ provides a uniquely clean demonstration of algebraic constraints, with all 239 quadruplets confined to exactly 2 out of 23 possible residue classes.

(v) The non-monotonicity of $L(k)$, controlled by the forbidden density $\rho(k) = (k-1)/p_0(k)$, reveals an intricate interplay between the growth of forbidden residues and the irregularity of the least prime in arithmetic progressions.

Future directions include extending the computation to larger ranges ($n \leq 10^{10}$ or beyond) to improve the statistics for rare constellations, investigating the secondary terms in the Bateman–Horn asymptotics, and studying $L(k)$ for composite exponents.

Acknowledgments

The computations were performed using Python with the `gmpy2` library for arbitrary-precision arithmetic and BPSW primality testing. Parallel processing was implemented using Python's `multiprocessing` module. The author thanks the developers of `gmpy2` and the GMP library for their excellent software. All source code, data, and figures are available at https://github.com/Ruqing1963/prime-constellations.

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